

The Effect of LIHTC on Housing Markets: Quantities, Rents, and Welfare

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We are interested in how LIHTC units affect housing markets.

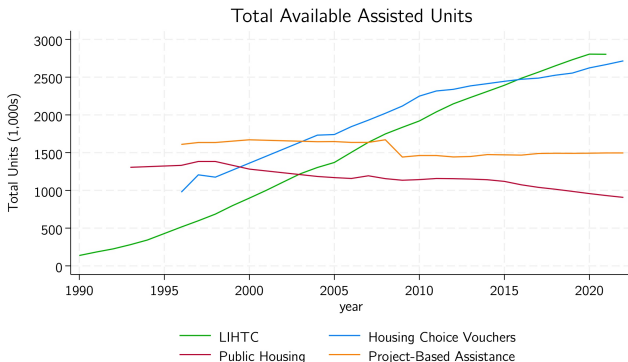
1. Does LIHTC increase the stock or rental housing?
2. Does LIHTC affect rent?
3. Does LIHTC have important unintended or external consequences?
4. How much surplus does LIHTC generate, and who gets it?

We estimate a blockgroup level housing market outcomes using a DiD organized around the blockgroups subject to winning and losing LIHTC applications.

We use estimated treatment effects from a LIHTC award/project as the basis for a simple welfare analysis.

US Housing programs

- ▶ Three classes of public housing support; public housing, Section 8 voucher programs, and developer based subsidy programs.
- ▶ LIHTC is a developer based subsidy program. There are about 2.8m LIHTC units in 2023. At 150,000\$ each, this is 420b\$ worth of apartments.



How LIHTC works I

Established by the 1986 Tax Reform Act, and run by the IRS

- ▶ Federal government awards LIHTC tax credits to states on a per person basis, 1.25\$/resident in 1986-2001 (except 0.93\$ in 1989). Increased to 1.75\$ and indexed to inflation in 2003.
- ▶ \$3.00 per person in 2025, with a minimum small population state allocation of \$3.5m.

Developers apply to state housing authorities for LIHTC tax credits with a detailed description of their project

- ▶ State housing authorities award LIHTC credits to applications in a once or twice annual process using a ranking scheme partly determined by the enabling legislation. Some of the criteria are subjective (e.g., sustainability).

How LIHTC works II

These tax credits are allocated over a 10 year period

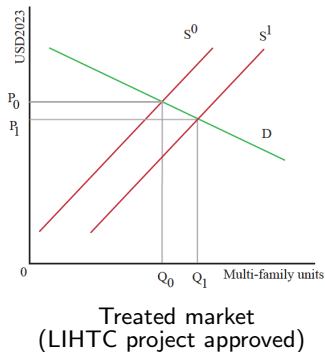
- ▶ In the case of a 9% LIHTC, for example, developers get 9% of a project's eligible development costs each year for 10 years.
- ▶ These credits are usually sold in a secondary market to finance construction costs. Worth about 70% of non-land costs.
- ▶ "Qualified Census Tracts" get a subsidy of 91% of capital.

Mostly built in poor neighborhoods

- ▶ A LIHTC project must rent 20% of units to households with income $< 50\%$ Area Median Income (AMI) or 40% of households with income $< 60\%$ AMI for 15 or 30 years.
- ▶ In general 100% of the units in a LIHTC building are designated as LIHTC units, to maximize tax credits.
- ▶ Rent is capped at 30% of 60% of AMI with some exceptions.

Welfare/Model I

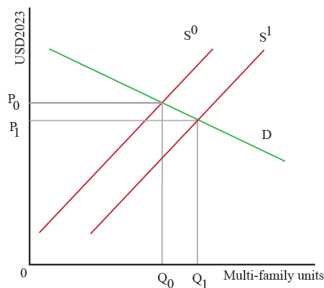
Start with equilibrium (P_0, Q_0) . Approval of a LIHTC project shifts the supply curve in the treated local market from Q_0 to Q_1 .



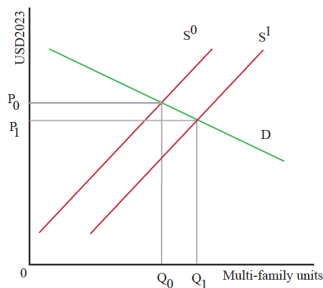
- ▶ In this example, LIHTC increases supply and lowers rent.
- ▶ Identifying restriction is that LIHTC doesn't shift demand. We estimate $(\Delta P, \Delta Q)$ and calculate change in surplus and transfers.

Welfare/Model II

We compare the treated market with a control market that had a rejected LIHTC project:



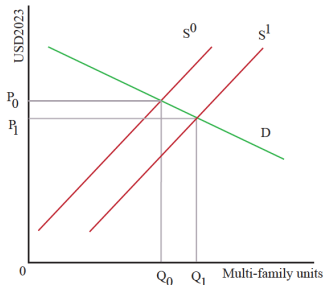
Treated market
(LIHTC project approved)



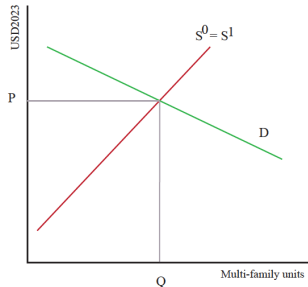
Control market
(LIHTC project rejected)

- In this example, there is **100% crowd-out** since developers in the control market would have constructed the same number of units in the absence of LIHTC tax credits.

Welfare/Model III



Treated market
(LIHTC project approved)

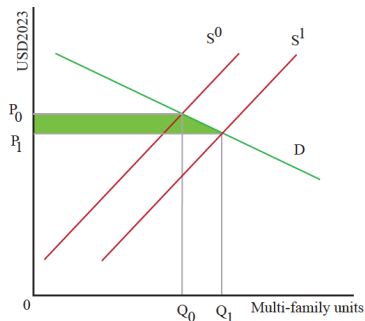


Control market
(LIHTC project rejected)

- In this second example, there is close to **0% crowd-out** since developers would *not* have constructed the new multifamily units in the control market in the absence of LIHTC tax credits.

Welfare/Model IV

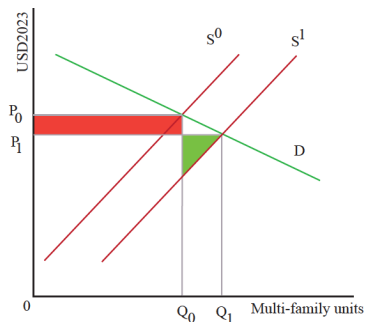
We will also estimate the consumer surplus gains from a LIHTC project:



- When market rents decline, both existing residents and LIHTC occupants benefit from the lower prices.

Welfare/Model V

The LIHTC shock also results in producer surplus losses and gains



- ▶ Lower market rents negatively impact existing landlords - transfers from landlords to existing renters.
- ▶ New construction benefits LIHTC developers.

Literature I

- ▶ 100% Crowd out $\implies$ 100 LIHTC units changes housing stock by zero units. 0% Crowd out $\implies$ 100 LIHTC units changes housing stock by 100 units.
 - ▶ Sinai and Waldfogel [2005]. 1990 Cross-section, county and MSA. 50-70% crowd-out for all public housing units.
 - ▶ Eriksen and Rosenthal [2010]. 1990-2000 first differences, MSA, county, 10k disk around a tract. OLS 15% crowd out, IV 100% crowd out for LIHTC.
 - ▶ Baum-Snow and Marion [2009] RDD on threshold for QCTs in 1990-2000. About 20% crowd out of rental units.
 - ▶ Soltas [2024]. Similar win/lose design used in this paper, but compares LIHTC projects at parcel level. 75% crowd out, but a little different definition.

Literature II

- ▶ Prices:
 - ▶ Baum-Snow and Marion [2009] find 10% increase in price of owner occupied units from 100 LIHTC units in block-group.
 - ▶ Diamond and McQuade [2019], zero average effect on owner occupied, -2%(+6%) in wealthy(poor) neighborhoods, decaying to 0% 1km from LIHTC project. Mcculloch [2025] reanalysis gets -2%.
 - ▶ Burge [2011], LIHTC $\approx$ 10% below hedonic price of market rate units in one city. Soltas [2024] LIHTC units 12% below comparable market rate units.
- ▶ Quality: Burge [2011] LIHTC units tend to be newer than market units (126 buildings in Tallahassee in 2002).
- ▶ Externalities: Baum-Snow and Marion [2009], decline in tract income, increase in share renters, and faster turnover, but this looks mechanical. Diamond and McQuade [2019] decreases in crime near LIHTC units in poor neighborhoods.

Literature III

Summary:

- ▶ LIHTC is hard to study because (1) outcomes are hard to observe (e.g. demographics and rent of LIHTC residents, locations of LIHTC projects) and, (2) unraveling the assignment process is difficult.
- ▶ From 25 years of research we conclude
 - ▶ Crowd-out is between 20% and 100%. The two most compelling research designs give 20% and 75%.
 - ▶ Prices of owner occupied housing near LIHTC change between -2% and $+10\%$.
 - ▶ LIHTC units are newer on average and rent for about 11% less than comparable market rate units, on average.(check)
- ▶ There has been surprisingly little interest in *market* level rental prices and housing consumption.

Data I

We use three main data sources

- ▶ Purpose collected and digitized lists of applications and awards from State Housing Authority websites and a commercial data aggregator (Application data). Each application reports state, year, successful or not, address, # units, rehab vs. new.
- ▶ Housing and Urban Development (HUD) data inventorying all LIHTC awards and units in service. Each award/project has a date, number of units, and geocoded latitude and longitude.
- ▶ Census/American Community Survey block group level data for 1990, 2000, 2005-2009, 2010-2014, 2015-2019, 2019-2024. We standardize to 2010 boundaries. This is where we get most of our outcomes.
- ▶ Also: CORELogic (parcel level taxes and transactions), National Housing preservation Database (NHPD), (competitor to the HUD data with substantial overlap).

Data II

We scraped lists of applications and awards from state housing authority websites and a commercial data aggregator

2016 Final HOME/Housing Credit Applications
ALABAMA HOUSING FINANCE AUTHORITY

Project Name	Project Address	Project City	Project ZIP	Project County	Owner Information	F Units	Tract Size	Project Type	Construction Status	Fin. Profile, 2015	2016 Housing Units	Status
Central Green	2401 Green Blvd	Mobile	36608	Mobile	Alabama Green LP 2401 Green Blvd Mobile, AL 36608	60	2.00	Medium	Under Construction	Fin Profile	10000	Construction Complete
Greenway Apartments	Greenway Blvd	Mobile	36608	Mobile	Greenway Housing LP 2401 Green Blvd Mobile, AL 36608	60	2.00	Medium	Under Construction	Fin Profile	10000	Construction Complete
Greenway Commons	Greenway Blvd	Mobile	36608	Mobile	Greenway Housing LP 2401 Green Blvd Mobile, AL 36608	60	2.00	Medium	Under Construction	Fin Profile	10000	Construction Complete
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1/14/2016

APPLICANTS SHOULD NOTIFY DATA MANAGER OF ANY CHANGES TO INFORMATION

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2016 HOME/Housing Credit Funding Approvals
ALABAMA HOUSING FINANCE AUTHORITY

Project Name	Project Address	Project City	Project ZIP	Project County	Owner Information	F Units	Tract Size	Project Type	Construction Status	Fin. Profile, 2015	2016 Housing Units	Status
Greenway Commons	Greenway Blvd	Mobile	36608	Mobile	Alabama Green LP 2401 Green Blvd Mobile, AL 36608	60	2.00	Medium	Under Construction	Fin Profile	10000	Construction Complete
Greenway Commons	Greenway Blvd	Mobile	36608	Mobile	Alabama Green LP 2401 Green Blvd Mobile, AL 36608	60	2.00	Medium	Under Construction	Fin Profile	10000	Construction Complete
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Greenway Commons	Greenway Blvd	Mobile	36608	Mobile	Alabama Green LP 2401 Green Blvd Mobile, AL 36608	60	2.00	Medium	Under Construction	Fin Profile	10000	Construction Complete

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Alabama, 2016 application

Alabama, 2016 awards

- Digitize application and award files.
- Match Applications to Awards.
- Geocode addresses.

Data III

Application data:

- ▶ All LIHTC since 1986: 50 states $\times$ 40 years of application rounds, 1 or 2 rounds per year.
- ▶ We collected lists of applications and awards for 479 application rounds in 43 states, mostly since 2000s

Sample	Applications	Approved projects	Rejected projects
Our full sample (43 states)	31,012	15,360	15,652
Post geocoding (38 states)	17,183	8,695	8,488
Treated in 2010-14 (23 states)	2,744	1,176	1,568

Data IV

Issues:

- ▶ Geocoding addresses is hard, and failure to geocode an address is the main way we lose applications. This occurs because we have an imprecise address (most often) or neither the census or google geocoder could resolve the address. We have two qualities of geocoding
 - ▶ Exact: "125 Apple Lane, Appleville, CA 90210".
 - ▶ Not exact: "Corner of Apple and Orange Rd, Appleville."

We sometimes restrict to the exact geocoding sample.

- ▶ Standardizing Census/ACS to constant 2010 boundaries introduces measurement error to ACS data that is not present in Application data (which is geocoded to 2010 boundaries). We sometimes restrict attention to block groups that experience at most very small changes in their boundaries over time.

Research design I

- ▶ Unit of observation is a block-group period.
- ▶ A period is a 5 (10) year ACS (Census) window; 1990, 2000, 2005-2009, 2010-2014, 2015-2019, 2019-2023. Six periods.
- ▶ Staggered adoption DiD. We control for state fixed effects in all estimations. We use the doubly robust estimator [Sant'Anna and Zhao, 2020].

Research design II

Award/ treatment path	1990-99	2000-4	2005-9	2010-14	2015-19	2020-4
P_0 (control)	0	0	0	0	0	0
P_{x1} (drop)	1	*	*	*	*	*
P_{x2}	0	0	0	0	0	1
P_{11} (main)	0	1	*	*	*	*
P_{12}	0	0	1	*	*	*
P_{13}	0	0	0	1	*	*
P_{14}	0	0	0	0	1	*
P_{21} (drop)	0	0	1	0	0	*
P_{22} (nice)	0	0	0	1	0	0

- ▶ Defining treatment is tricky. Black 1/0 describes award path (from application data), shading shows treatment.
- ▶ Pink: control blockgroups, application(s), but no award ever.
- ▶ Gray; drop, no pre- or post-award data.
- ▶ Green; award and treatment coincide, P_{21} is financial crisis. 'Nice sample' is award path P_{22} .
- ▶ Blue; main sample. Treatment and award don't coincide, but big sample.

Research design III

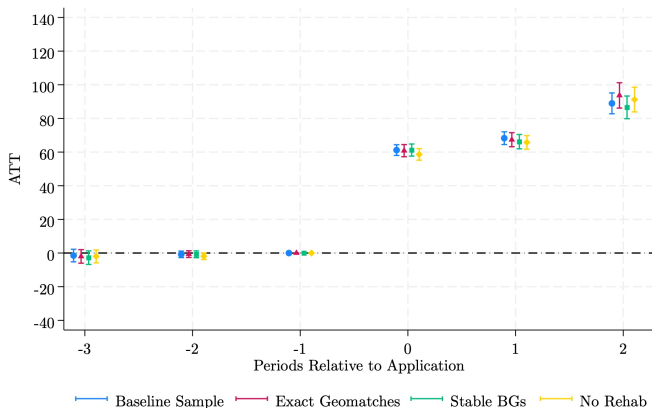
Why block-groups?

- ▶ We want to describe a 'housing market'.
- ▶ Older studies use larger units for this purpose, counties or MSAs.
- ▶ Recent papers show that housing spillovers operate over smaller spatial scales; Baum-Snow and Marion [2009], Diamond and McQuade [2019], Pennington [2021] and Asquith et al. [2023] $\implies$ housing markets are small.
- ▶ We check for treatment effects (spillovers) in nearby block-groups.

Treated block-groups (regardless of sample) are spatially separate from controls.

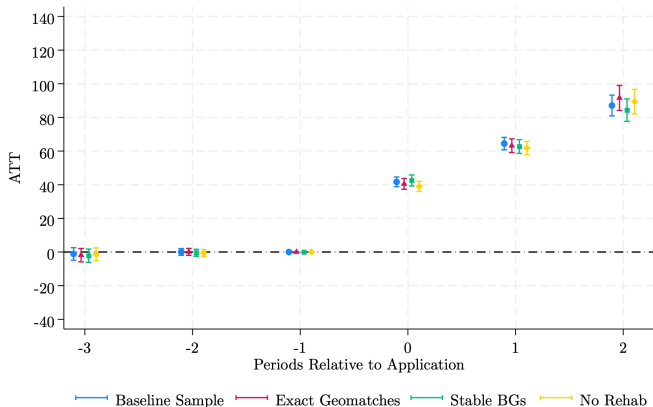
Treated and control block groups (regardless of sample) are not balanced on levels; treated block groups are a little poorer. Identification comes from parallel trends.

Main sample, HUD units awarded



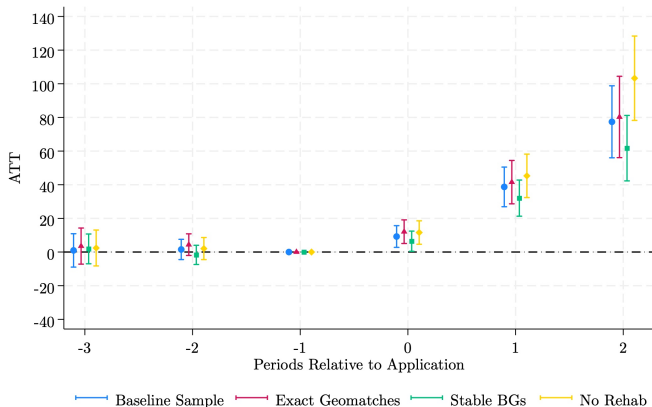
- (1) Developers follow up on state awards so HUD awards happen with awards in our application data. (2) The height is important ≈ 77 units. (3) Some units have complicated award paths.

Main sample, HUD units Placed in service.



(1) There is a construction lag between award and placed in service. (2) 2-3 ACS periods post award, PIS and Awards converge
⇒ all awarded units are built.

Main sample, ACS Multi-family units.



(1) There is a lag before HUD units placed in service show up in the ACS (2) 2-3 ACS periods post award, ACS Multifamily converges to HUD units awarded.

Crowd out?

VARIABLES	(1) LIHTC Units Allocated (HUD)	(2) LIHTC Units Placed in Service (HUD)	(3) Total Multifamily Housing Units (ACS)
T-3	-1.498 (1.912)	-1.138 (1.904)	0.960 (5.060)
T-2	-0.780 (0.986)	0.034 (1.026)	1.533 (3.074)
T-1	0 (0)	0 (0)	0 (0)
T	61.180*** (1.629)	41.685*** (1.469)	9.219*** (3.291)
T+1	68.254*** (1.937)	64.466*** (1.881)	38.697*** (6.012)
T+2	88.954*** (3.154)	87.079*** (3.146)	77.400*** (10.916)
State FE	✓	✓	✓
Ctrl Grp Avg	29.5	27.2	208.5
Observations	42168	42168	42168

Callaway Sant'anna standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

89 LIHTC units awarded $\Rightarrow$ 87 placed in service $\Rightarrow$ 77 multi-family units $\Rightarrow$ $1 - \frac{77}{89} = 13\%$ CI95[-6%, 28%] crowd-out.

nice sample

Other housing outcomes

Spillovers

To estimate the effect of LIHTC units on housing markets in block-groups in the same tract as a treated block-group, define block-group as

- ▶ Treated with a LIHTC spillover if it is in the same tract as a treated block-group.
- ▶ Control for LIHTC spillovers if it is in the same tract as block-group a control blockgroup.
- ▶ Drop a few blockgroups that satisfy both conditions.
- ▶ On average, a treated blockgroup is in a tract with 1.5 other blockgroups.
- ▶ Spillover ATT is about 20 Multifamily units in main sample. This drops to 6 if we restrict attention to stable borders. Spillover multifamily units look like they are mostly measurement error.

Summary

- ▶ Our main sample suggests about 13% crowd-out if we measure units with ACS Multifamily.
- ▶ Other ACS measures of housing units look about the same.
- ▶ Spillovers are small in a sample with stable block-groups.
- ▶ Easiest story is that there is no crowd out and the small spillovers reflect measurement errors; one LIHTC unit gives one multifamily unit in the targeted blockgroup.

Average Award Impulse for Treated Units, Example I

$i \sim$ blockgroup, $D \sim$ treatment, $g \sim$ treatment period, $t \sim$ period,
 $k \in \{1, 2\} \sim$ treatment index, $N_k \sim$ count of i in treatment group k .

$$k = 1 \quad P_1 = (0, 0, 1, 0, 0, 0, 0)$$

$$D_1 = (0, 0, 1, 1, 1, 1, 1)$$

$$k = 2 \quad P_2 = (0, 0, 1, 0, 1, 0, 0)$$

$$D_2 = (0, 0, 1, 1, 1, 1, 1)$$

Want *Average Award Impulse*. That is, treatment effects at $t = 3, 4, 5$ when all units receive D_1 .

Main idea: P_1 is 'simple award path'. $P_2 = P_1 + L^{-2}P_1$ is a 'compound award path'. Require treatment effects for D_1 and D_2 to satisfy the same relationship (for units receiving D_2).

For units actually receiving D_1 , estimate

$$ATT(g = 3, t, D_1, k = 1), \quad t = 3, 4, 5$$

Average Award Impulse for Treated Units, Example II

$$k = 1 \quad P_1 = (0, 0, 1, 0, 0, 0, 0)$$

$$D_1 = (0, 0, 1, 1, 1, 1, 1)$$

$$k = 2 \quad P_2 = (0, 0, 1, 0, 1, 0, 0)$$

$$D_2 = (0, 0, 1, 1, 1, 1, 1)$$

For units actually receiving D_2 , at periods $t = 3, 4$

$$A0 : ATT(g = 3, t, D_2, k = 2) = ATT(g = 3, t, D_1, k = 2),$$

Because the award paths for D_1 and D_2 agree in their first two post award periods.

Average Award Impulse for Treated Units, Example III

$$k = 1 \quad P_1 = (0, 0, 1, 0, 0, 0, 0)$$

$$D_1 = (0, 0, 1, 1, 1, 1, 1)$$

$$k = 2 \quad P_2 = (0, 0, 1, 0, 1, 0, 0)$$

$$D_2 = (0, 0, 1, 1, 1, 1, 1)$$

What about units receiving D_2 , at period $t = 5$?

With $P_2 = P_1 + L^{-2}P_1$, require 'additivity' and 'stationarity'

$$\begin{aligned} A1 : ATT(g = 3, t = 5, D_2, k = 2) &= ATT(g = 3, t = 5, D_1, k = 2) \\ &\quad + ATT(g = 5, t = 5, D_1, k = 2) \end{aligned}$$

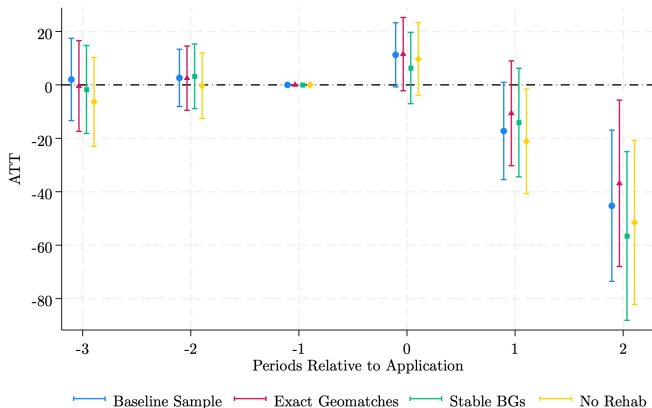
$$A2 : ATT(g = 3, t = 3, D_1, k = 2) = ATT(g = 5, t = 5, D_1, k = 2).$$

Together, $A0, A1, A2 \implies$

$$\begin{aligned} ATT(g = 3, t = 5, D_1, k = 2) &= ATT(g = 3, t = 5, D_2, k = 2) \\ &\quad - ATT(g = 3, t = 3, D_2, k = 2). \end{aligned}$$

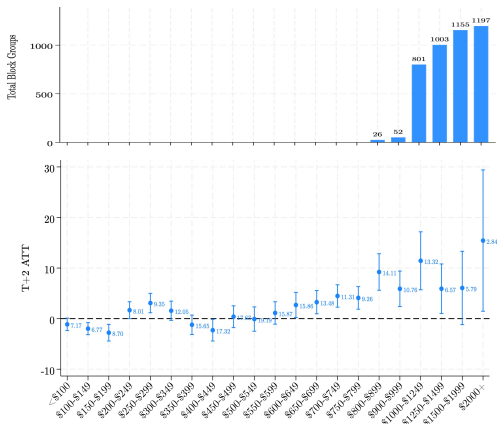
Both RHS terms are easy to estimate. Weight to get sample estimates. Bootstrap for standard errors.

Main sample, ACS Average Contract Rent.



Average contract rent declines by about 45\$ month per unit in treated block groups.

Main sample, change in rental units by initial rent.



Top: Histogram of blockgroups by LIHTC cap. Bottom: change in total rental units by rent bin in treated blockgroup. Much of the increase in units comes far from the LIHTC caps. Units for the very poor are not affected.

Prices, main sample

VARIABLES	(1) Avg Contract Rent	(2) Log(Avg Contract Rent)	(3) Avg Gross Rent	(4) Log(Avg Gross Rent)
T-3	2.025 (7.861)	0.002 (0.007)	4.541 (8.518)	0.004 (0.006)
T-2	2.620 (5.470)	0.004 (0.005)	2.170 (6.083)	0.004 (0.005)
T-1	0 (0)	0 (0)	0 (0)	0 (0)
T	11.267* (6.124)	0.013** (0.005)	10.843 (6.697)	0.013*** (0.005)
T+1	-17.248* (9.290)	-0.004 (0.007)	-25.698*** (9.518)	-0.005 (0.006)
T+2	-45.253*** (14.443)	-0.020* (0.011)	-73.118*** (15.291)	-0.025** (0.010)
State FE	✓	✓	✓	✓
Ctrl Grp Avg	872.8	6.7	1093.4	6.9
Observations	40085	39888	39074	39074

Callaway Sant'anna standard errors in parentheses. All prices are in 2024 dollars.

*** p<0.01, ** p<0.05, * p<0.1

- ▶ Rent ↓ 2.5% for average rental unit in treated block-group.
- ▶ Increase in 'affordability', rent < 30% income, but small compared to new multifamily units. Data? Turnover

Externalities/Demand Shifters

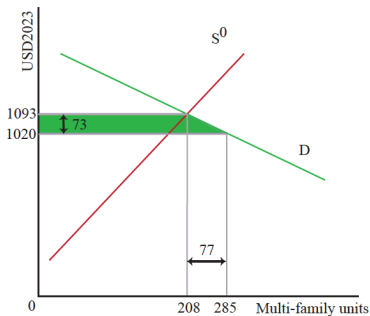
- ▶ Average rooms for renter households increase by 0.2. $T - 1$ control group mean 4.4. No effect on owner occupied or share of crowded units. Quality
- ▶ Owner occupied moved last five years at $T + 2$ up 8. $T - 1$ control group mean 158. For renters at $T + 1$, 25 of 216. There is a lot of turnover. LIHTC increases it a bit, some is mechanical Turnover
- ▶ Tiny positive effect on owner occupied price in ACS and CORE Logic at T . Nothing at $T + 1$, $T + 2$. Owner occupied
- ▶ At $T + 2$ treated blockgroups have 0.006 larger share of residents with 'some college'. $T - 1$ Control group mean is 0.3. No effect on income. Demographics

We can't prove that LIHTC awards do not shift demand, but if it caused a big shift, we would expect to see evidence of it in these outcomes.

Summary

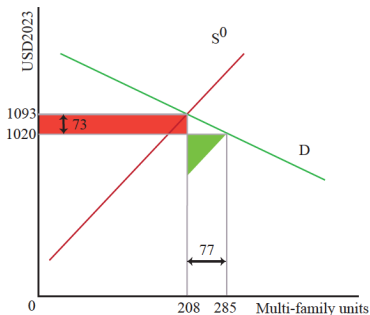
- ▶ An average treated block group experiences a 77 unit increase and a 73\$ rent decrease for all $208 + 77$ rental units in a block-group.
- ▶ Some of this may reflect changes in composition, but we can't see it in the ACS.
- ▶ Prices of owner occupied units probably doesn't change.
- ▶ Turnover rate is high. Probably a small increase from LIHTC, some is mechanical.
- ▶ Tiny changes in demographics.
- ▶ Spillovers look like measurement error.
- ▶ None of these findings suggests external effects from LIHTC.
- ▶ Probably small increase in share of households $< 50\%$ AMI paying $< 30\%$ income in rent.

Welfare I



- Green is Δ Consumer Surplus per tract from an average increase of 77 LIHTC units = 17,795\$/Month. 30yrPV at 10% is 2.04m\$. This is 26,436\$ per unit.

Welfare II



- ▶ Red box gives rent decrease for infra-marginal 208 units, $208 \times 73 = 15,184$ month.
- ▶ 30yrPV at 10% is 1.72m\$.
- ▶ This means a 22,307\$ transfer from landlords to tenants for each of the 77 marginal units.

Welfare III

Producer surplus (green triangle)

- ▶ Rent on marginal units: 77 units, at 1020\$/mo is $1020 \times 77 = \$78,540/\text{mo}$. 30yrPV at 10% is \$8.84m, or \$115,385/unit.
- ▶ Developer costs: Ward and Schlake [2025] estimate mean cost of a LIHTC unit: \$85k in Texas, \$94k in Colorado, and \$400k in California. Soltas [2024] has \$145k application mean. We use \$150k.
- ▶ Producer surplus is the difference; loss of \$34,615/unit
- ▶ LIHTC subsidy is about 70% of capital. Capital is about 85% of cost per unit $= 150\text{k} \times 0.85 \times 0.7 = \$89,250$ per unit.

Total per unit:

$$\begin{aligned} & -\$34,615 \text{ producers surplus} + \$26,436 \text{ consumers surplus} \\ & \quad -\$22,307 \text{ transfer from landlords} = -\$30,486 \end{aligned}$$

Welfare IV

LIHTC subsidies of \$89,250 per unit create:

- ▶ Total surplus of -\$30,486 per unit.
- ▶ Transfer from landlords to tenants of \$22,307.
- ▶ increase in consumers surplus of \$26,436.
- ▶ Developers lose \$34,615 without the \$89,250 subsidy.
- ▶ $\implies$ households would have rather the subsidy than the housing. Developers are better with the subsidy. Landlords probably don't want the program.
- ▶ Sensitive to cost of construction and discount rate.

More on incidence

How much of rent decrease is due to lower rents in LIHTC units?

We don't know, and there is no data on actual rents paid on these units.

- ▶ Sample Mean AMI is 90k $\implies$ LIHTC rent cap \$1,125/mo $>$ \$1,093/mo. On average, LIHTC rent cap is not binding.
- ▶ Soltas [2024] uses rents claimed on applications to estimate that LIHTC rents are 12% below market.
- ▶ If the LIHTC cap does not bind, and 77 new LIHTC units rent for a 12% discount from control mean 1093\$ month, then average rent should fall by about 35\$/month. Together with evidence that rents fall for units far from the LIHTC caps $\implies$ most of the drop in average prices is a supply response, not mechanical.
- ▶ This is pure redistribution.

Conclude

We investigate the effect of LIHTC on housing markets. We find

- ▶ A marginal LIHTC unit warranted adds about one multi-family rental unit to the housing stock. There is essentially zero crowd out.
- ▶ An average LIHTC project reduces rental prices by about \$73/month. No effect on owner occupied market.
- ▶ Spillovers of development to adjacent block groups look like measurement error.
- ▶ Effects of LIHTC on composition/demographics appear small. No obvious evidence of large external effects.
- ▶ On average, LIHTC transfers \$89k to developers to increase tenant welfare by \$26k, decrease incumbent landlord surplus by \$22k, and destroy \$30k of surplus.

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Nice Sample

VARIABLES	(1) LIHTC Units Allocated (HUD)	(2) LIHTC Units Placed in Service (HUD)	(3) Total Multifamily Housing Units (ACS)
1990	2.436 (4.659)	4.562 (4.721)	-3.537 (10.643)
2000	-1.647 (2.933)	0.436 (3.239)	3.956 (8.590)
2005-2009	0 (0)	0 (0)	0 (0)
2010-2014	55.122*** (3.204)	44.230*** (3.097)	-0.552 (9.202)
2015-2019	54.055*** (3.495)	55.002*** (3.457)	24.350* (13.003)
2020-2024	53.210*** (3.654)	54.028*** (3.721)	29.074 (19.179)
State FE	✓	✓	✓
Ctrl Grp Avg	34.3	32.7	218.5
Observations	11862	11862	11862

Callaway Sant'anna standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Much noisier for units with awards only in 2010-2014. We can't reject anything interesting. [back](#)

Related housing supply outcomes

VARIABLES	(1) Total Housing Units	(2) Total Vacant Units	(3) Total Renter-Occ. Units	(4) Total Owner-Occ. Units	(5) Total Multifamily Units	(6) Total Single Family Units	(7) Total Multifamily Renter-Occ. Units	(8) Total Multifamily Owner-Occ. Units	(9) Total Single Family Renter-Occ. Units	(10) Total Single Family Owner-Occ. Units
T-3	13.030 (10.941)	2.638 (2.090)	5.712 (4.646)	5.139 (7.172)	0.960 (5.060)	9.479 (8.733)	3.980 (4.028)	-1.114 (0.970)	2.006 (1.913)	3.708 (6.891)
T-2	7.940 (5.788)	1.660 (1.358)	3.689 (2.894)	2.475 (3.761)	1.533 (3.074)	5.295 (4.488)	3.221 (2.552)	-0.732 (0.619)	1.491 (1.261)	2.786 (3.618)
T-1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
T	4.584 (6.665)	6.507*** (1.802)	4.465 (3.305)	-5.386 (4.430)	9.219*** (3.291)	-7.879 (5.238)	7.066** (2.877)	1.185* (0.695)	-1.724 (1.457)	-8.188* (4.273)
T+1	37.271*** (12.680)	5.459*** (1.829)	33.736*** (5.841)	-1.401 (8.342)	38.697*** (6.012)	-0.308 (9.465)	34.036*** (5.050)	2.271** (0.978)	1.006 (2.082)	-4.387 (8.034)
T+2	74.153*** (22.285)	8.968*** (2.852)	65.547*** (10.295)	0.536 (14.143)	77.400*** (10.916)	-0.833 (16.051)	67.519*** (8.938)	5.590*** (1.837)	4.377 (3.286)	-7.549 (13.587)
State FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ctrl Grp Avg '05-'09	681.0	83.2	254.0	344.4	208.5	425.1	163.4	12.3	77.3	305.8
Observations	42168	42168	42168	42168	42168	42168	42168	42168	42168	42168

*** p<0.01, ** p<0.05, * p<0.1

LIHTC awards increase the stock of Multifamily/rental units by about one unit for each Awarded Unit recorded by HUD. They have no effect on the single family housing stock. [back](#)

Spillovers

SAMPLE VARIABLES	Baseline	Neighboring Block Groups		Neighboring Block Groups (Stable BGs Only)	
	(1)	(2)	(3)	(4)	(5)
	Total Single Family Units (ACS)	LIHTC Units Placed in Service (HUD)	LIHTC Units Total Multifamily Housing Units (ACS)	LIHTC Units Placed in Service (HUD)	LIHTC Units Total Multifamily Housing Units (ACS)
T-3	9.479 (8.733)	0.337 (0.999)	-0.937 (2.404)	1.092 (1.533)	-5.263 (3.679)
T-2	5.295 (4.488)	-0.225 (0.595)	-1.482 (1.491)	-0.051 (1.089)	-0.732 (2.454)
T-1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
T	-7.879 (5.238)	2.500*** (0.621)	5.122*** (1.586)	-0.651 (0.687)	-2.227 (1.541)
T+1	-0.308 (9.465)	3.589*** (0.928)	9.360*** (2.568)	2.474*** (0.692)	3.435** (1.542)
T+2	-0.833 (16.051)	5.393*** (1.595)	20.326*** (4.847)	2.878*** (1.040)	6.859*** (2.426)
State FE	✓	✓	✓	✓	✓
Ctrl Grp Avg in '05-'09	425.1	12.1	141.9	11.1	133.1
Observations	42168	84594	84594	64632	64632

On average a treated block-group spills over to 1.5 other block-groups in the same tract. Spillovers look like they are mostly measurement error. [back](#)

"Affordability"

VARIABLES	(1) Hshlds Below AMI (Underest.)	(2) Hshlds Below AMI (Overest.)	(3) Hshlds Paying <30% of Income on Rent	(4) Hshlds Below AMI Paying <30% of Income on Rent (Underest.)	(5) Hshlds Below AMI Paying <30% of Income on Rent (Overest.)
T-3	8.507* (4.936)	11.003* (5.854)	3.136 (2.812)	3.361** (1.715)	2.157 (2.405)
T-2	2.955 (3.044)	4.177 (3.663)	4.257** (1.831)	3.192*** (1.227)	3.221** (1.589)
T-1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
T	-2.597 (3.113)	-1.841 (3.706)	3.539* (2.128)	0.507 (1.357)	2.306 (1.885)
T+1	23.977*** (5.675)	23.507*** (6.846)	8.345*** (3.181)	3.858** (1.630)	4.773 (2.932)
T+2	48.149*** (9.701)	50.570*** (11.858)	21.838*** (5.430)	8.029*** (2.515)	14.929*** (5.014)
State FE	✓	✓	✓	✓	✓
Ctrl Grp Avg	375.2	427.4	116.5	68.7	94.7
Observations	42168	42168	42168	42168	42168

Callaway Sant'anna standard errors in parentheses.

AMI held constant at 2000 level.

*** p<0.01, ** p<0.05, * p<0.1

No obvious increase in housing affordability in LIHTC treated block-groups. [back](#)

Quality/composition

VARIABLES	(1) Total Crowded Units	(2) Avg Rooms: Renter Hshlds	(3) Avg Rooms: Owner Hshlds
T-3	1.774 (1.416)	0.022 (0.029)	-0.044 (0.041)
T-2	0.138 (0.971)	0.117*** (0.022)	-0.035 (0.026)
T-1	0 (0)	0 (0)	0 (0)
T	-2.861*** (1.029)	0.141*** (0.024)	-0.028 (0.019)
T+1	-0.004 (1.451)	0.057** (0.025)	-0.041 (0.029)
T+2	1.809 (2.384)	0.197*** (0.041)	0.007 (0.036)
State FE	✓	✓	✓
Ctrl Grp Avg	20.6	4.4	6.1
Observations	42161	37771	37373

Callaway Sant'anna standard errors in parentheses.

*** p<0.01, ** p<0.05, * p<0.1

Maybe a small effect on the size of rental units. If rooms decreases by 0.15 from base of 4.1, from adding 51 units to a stock of 264, then new units are about 3.2 rooms. [back](#)

Externalities, turnover

VARIABLES	(1)	(2)	(3)	(4)
	Total	Total	Total	Total
	Renter Hshlds Moved in	Renter Hshlds Moved in	Owner Hshlds Moved in	Owner Hshlds Moved in
	≤ 1 Yr Ago	≤ 5 Yrs Ago	≤ 1 Yr Ago	≤ 5 Yrs Ago
T-3	2.290 (3.331)	2.796 (4.544)	1.630 (2.626)	6.862 (6.559)
T-2	3.517* (2.091)	4.087 (2.841)	-0.788 (1.387)	1.886 (3.389)
T-1	0 (0)	0 (0)	0 (0)	0 (0)
T	6.464*** (2.187)	5.244* (3.034)	-0.279 (1.295)	0.743 (2.947)
T+1	12.720*** (2.788)	25.129*** (4.556)	2.468 (1.583)	2.478 (3.796)
T+2	-1.339 (3.868)	8.355 (7.190)	0.651 (2.191)	8.297** (4.212)
State FE	✓	✓	✓	✓
Ctrl Grp Avg	141.6	216.0	64.9	158.5
Observations	42168	42168	42168	42168

Callaway Sant'anna standard errors in parentheses.

*** p<0.01, ** p<0.05, * p<0.1

We do not see an increase in turnover, though the baseline turnover rate is very high. Recall, 264 rental and 350 owner occupied units. This does not look like an important externality at work. [back](#)

Externalities, owner occupied prices

VARIABLES	(1) Avg Sale Price (CoreLogic)	(2) Log(Avg Sale Price) (CoreLogic)	(3) Price Index (CoreLogic+)	(4) Avg Value (ACS)	(5) Log Avg Value (ACS)
T-3	.	.	.	2113.31 (4067.69)	0.009 (0.010)
T-2	-2829 (5582)	-0.006 (0.009)	-1.14 (1.36)	1787.31 (3833.17)	0.004 (0.009)
T-1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
T	9542* (4986)	0.026*** (0.008)	3.02*** (1.01)	6897.31* (3938.20)	0.012 (0.008)
T+1	6812 (6617)	-0.004 (0.009)	0.69 (2.15)	9695.36 (9647.01)	0.031** (0.016)
T+2	7731 (9603)	0.011 (0.012)	-2.61 (3.12)	2204.32 (13334.55)	0.039 (0.025)
State FE	✓	✓	✓	✓	✓
Ctrl Grp Avg	259423.21	12.232	113.76	261539.28	12.29
Observations	26556	26556	17149	27245	27245

Callaway Sant'anna standard errors in parentheses.
 All prices are in 2024 dollars, adjusted using CPI-U-RS.
 *** p<0.01, ** p<0.05, * p<0.1

We do not see changes in the prices of owner occupied units.

[back](#)

Demographics I

VARIABLES	(1) Avg Household Income	(2) Log Avg Household Income	(3) Share of Pop 25+ with HS Degree, No College	(4) Share of Pop 25+ with Some College, No Bachelor's	(5) Share of Pop 25+ with Bachelor's
T-3	-1327.194* (710.774)	-0.010 (0.008)	-0.009*** (0.003)	0.004 (0.003)	-0.005 (0.003)
T-2	-1315.384*** (486.721)	-0.017*** (0.006)	-0.005** (0.002)	0.001 (0.002)	-0.001 (0.002)
T-1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
T	834.712 (535.955)	0.006 (0.006)	-0.005** (0.002)	0.006*** (0.002)	0.004** (0.002)
T+1	-652.243 (737.121)	-0.007 (0.008)	-0.004 (0.003)	0.006** (0.003)	0.003 (0.003)
T+2	-570.440 (1160.595)	0.004 (0.012)	-0.001 (0.004)	0.015*** (0.004)	0.002 (0.004)
State FE	✓	✓	✓	✓	✓
Ctrl Grp Avg	77362.6	11.2	0.3	0.3	0.2
Observations	41732	41732	42110	42110	42110

Callaway Sant'anna standard errors in parentheses.

All prices are in 2024 dollars, adjusted using CPI-U-RS.

*** p<0.01, ** p<0.05, * p<0.1

No measurable changes in demographics. [back](#)